

Explanatory Construction Semantics 2.0

A structural model class for explanatory attributions

Version: ECS 2.0. **Relation to 1.0:** an audit-revised successor, not a new foundation. Every repaired item cites the audit finding (F1–F13) and recommendation (R1–R8) of ECS_1_0_Audit.md that motivates it.

What this document is. A class of relational structures for stating, and thereby criticizing, attributions of explanation, reason-use, understanding, provenance, repair, knowledge creation, and universality. It is not a semantics in the logician’s sense (F11): it assigns no compositional meanings to a language. It is a satisfaction-condition framework — it says what would have to be true for an attribution to hold, not whether it holds (F1). It is not a theory of which explanations are true, which beings are creative, or which systems are universal. It makes no claim that any actual person, machine, or institution belongs to any of its classes.

Constitutive conjecture (restated, with its status made explicit). The structural conditions defined below, correctly interpreted, capture all and only genuine explanatory accounting. This conjecture has the epistemic status of a conceptual analysis in the style of the Gettier literature: it stands or falls against pre-theoretic verdicts on cases, and its known failure mode is absorption of counterexamples by post-hoc re-graining. Section 9.4 therefore imposes a grain-fixing discipline (R6) as part of the framework itself, not as an optional virtue. A candidate satisfying every condition that is nevertheless no explanation refutes sufficiency; a genuine explanation that cannot satisfy them under any grain fixed in advance refutes necessity. Neither verdict is available to a judge with no independent grip on the cases; the framework does not pretend otherwise (F8).

Part 0 — The Primitive Load Registry (R1)

Every item below is **declared model data**: the framework takes it as input and never derives it. The registry exists so that no application can silently inherit a burden it has not discharged. Each entry lists what the primitive must supply, what would falsify a proposed supply of it, and which defined predicates become vacuous if it is absent or wrong.

#	Primitive	Must supply	A proposed supply is falsified by	Predicates that collapse without it
P1	Representation $\text{Rep}_{\{\beta, \ell\}}(o, c, h)$	Which occurrences carry which contents, at boundary and grain , preserving bindings and transformations	A content-preserving recoding that changes the attributed transitions, or a content change that leaves them invariant	Understanding, reason-use, (G), (EK), (U1)–(U3)

Table 1 – continued

#	Primitive	Must supply	A proposed supply is falsified by	Predicates that collapse without it
P2	Physical module θ	Substrate state spaces, allowed histories, controlled actions, resources, disturbance	A claimed controlled action shown not to be physically admitted, or a resource shown to be inexhaustible	(CT1)–(CT4), Can , CanAdv , every capability attribution
P3	Role interpretation of components	contracts, Approx_θ , $\text{PhysicallyCon-structible}_\theta$	contrary to the contract	
P3	Role interpretation of components	Which target components are causal assignments, measurements, constitutive rules, and what edits mean	An “intervention” shown to be an observation equation added alongside an incompatible retained assignment	Anchoring clause of (E); hence everything built on (E)
P4	Normative relation $\square$ (when invoked)	Which reasons bear on which works in which aesthetic or normative respects	Showing the asserted reason-relation reduces to approval or enactment	Objective aesthetic attributions only
P5	Enabling predicate Enable and admissible	Physically and semantically meaningful resources, stated independently of the success claim	A condition whose only description is “conditions under which the system succeeds,” or one delivering the target achievement itself (see 8.3)	(U1), (U2)
P6	Domains $\square_\theta$, $\square^{\text{adv}}_\theta$	The subject matters and coherently posed challenges, specified independently of the candidate’s record	A domain shown to have been carved around observed successes or failures	(U3)
P7	Grain and boundary	Which structural distinctions count as differences, and where the attributed system ends	A grain changed after contrast evidence was examined (Section 9.4)	Every indexed predicate

Rule R1-1. Any attribution stated in this framework must name the instances of P1–P7 it relies on. An attribution with an unnamed dependency is incomplete, not false; it becomes false only when a named supply is falsified.

Part 1 — Organizations and questions

1.1 Organizations (unchanged from 1.0)

An organization is a structure

[$D=(V,(X_v)_{v \in V},J,B,A,L,role).$]

(V) is a set of ports; each (X_v) is a nonempty value domain; a valuation is an element of $(X_v \rightarrow V)$. (J) indexes components with footprints $(V_j \rightarrow V)$. (B) contains boundary conditions. (A) contains admitted edits and their composable sequences, including identity (1), associative where defined, with stated typing. For each (j,a,b) the interpretation supplies

[$L_j(a,b) \subseteq \{v \in V_j \mid X_v \in \text{Sol } D(a,b) = \{z \mid X_z D: j, z/\{V_j\} \in L_j(a,b)\}\}.$]

Deleted components impose the full relation on their ports. An intervention setting a variable replaces its assignment. Values may be paths, functions, fields, or histories; (O) imposes no finiteness, acyclicity, or linearity. Multiple solutions remain multiple.

1.2 Questions (unchanged in substance)

A question is $(p=(D, (b_0), C, Q, O_p))$ with scope $(\cdot)$, baseline $((1, b_0))$, respect $(\cdot)$, contrast contract (C), query operator (Q), answer profile

[$\text{Ans}_p(a,b) = Q(D,a,b),$]

and obligations (O_p) . A question may be mistakenly posed; exposing that is another question. The contract is criticizable and replaceable; what is prohibited is changing it during an assessment without recording the change.

1.3 Contents, occurrences, grains

Occurrences are located carriers; contents are interpreted organizations. The representation relation $\text{Rep}_{\{\beta, \ell\}}$ is primitive P1. A grain β fixes which structural distinctions count as differences. A genuine recoding is an isomorphism of retained structure at β ; coarsening is not automatically an isomorphism. Representation does not require successful explanation: a system can represent a false theory, misidentify a target, or understand an invalid objection.

Part 2 — Accounting

2.1 The typed restriction algebra (R4; repairs F4)

Restriction is no longer an unanalyzed operation. Fix an interpreted candidate with commitment set $(\cdot)$, where each $(\cdot)$ is typed as exactly one of:

- **premise** $(\wedge P)$: an asserted content in a stated role;
- **rule** $(\wedge R)$: a component relation or constitutive rule;
- **binding** $(\wedge B)$: a connection between represented parts (an entry of a map, a role assignment, an anchor);

- **map** ($\hat{M}$): a translation $(\cdot, \cdot, \cdot)$ or a restriction of one.

For (W) , the restriction $(E|W)$ is defined by type:

1. **Premise removal** deletes the premise from its role. It does not assert its negation.
2. **Rule removal** replaces the component relation by the full relation on its ports, as in (O) .
3. **Binding removal** deletes the connection. Any commitment whose stated role references the removed binding is marked **dangling**; a dangling commitment is retained in the record but is inert — it contributes to no support until its reference is re-bound or its role is revised. Dangling is a status, not a deletion.
4. **Map removal** truncates the map's domain; any composition requiring the truncated segment is undefined, and claims depending on that composition lapse. Partial truncation of a map's codomain is a new map, not a restriction of the old one.

A restriction is **well-formed** when its typing is declared. $(S_{\{E,p\}})$ below is defined only over well-formed restrictions. This algebra is a substantive choice: a criticism that shows a real inquiry cannot be typed this way restricts the application of $(S)-(M2)$, not their conditional truth.

2.2 Structural accounts (unchanged in form; R7 threading fix)

A structural account is $(E=(E,p, \cdot, \cdot, \cdot))$ with declared domains, $(\cdot)$ sending target valuations to explanatory valuations (possibly many-to-one), $(\cdot)$ translating edits and boundaries, $(\cdot)$ anchoring explanatory components to target components or derived subnetworks evaluated by (O) . The anchoring clause requires kind-typing: causal to causal, measurement to measurement, rule to rule — where the kinds of target components are supplied by primitive P3 and are thereby exposed as a named dependency of every (E) attribution (F7).

The conditions:

Structural fidelity. For comparisons in (C) ,

$$[\text{Sol}_D(a, b)] = \text{Sol}_{\text{E}}(\tau(a), \sigma(b)).$$

Composition fidelity. $(\cdot(1)=1)$ and $(\cdot(a_2a_1)=(a_2)(a_1))$ wherever composition is claimed. (C)

Question fidelity. $(\text{Ans}_{\text{E}}(\tau(a), \sigma(b)) = \text{Ans}_{\text{p}}(a, b)), \text{including the respect}(\kappa).(A)$

Non-circular dependence. The target answer is not an unanalysed boundary input; at least one admitted contrast removing or changing a nonempty block of active commitments changes or undetermines the answer profile in the claimed way, with all other declared boundary conditions preserved.

Non-vacuity. The baseline has a compatible state; a contrast family of mere notational variants does not satisfy non-circular dependence.

2.3 Graded accounting (R5; repairs F5)

Real attributions are partial. For a declared contrast subfamily (C) containing the baseline, define

[Account

2.4 What (E) excludes, and what it does not (unchanged)

An observed answer table without anchoring fails (E) ; a table that genuinely encodes the organization under its interventions may pass. “(p) because (p)” fails non-circular dependence. A true mechanism guessed on bad grounds can satisfy (E) while the guesser's support is circular; circular support for

acceptance is a different defect from circular explanatory structure. Depth is question-relative; (E) is not a measure of elegance. Explanatory direction is respect-relative, not globally asymmetric.

Part 3 — Supports, contribution, variation

3.1 Support families (restriction now typed)

Fix an interpreted candidate, question, background, and the typed restriction algebra of 2.2. For well-formed (W) ,

$$[S_{_}\{E,p\} = \{W : \text{Account}(E|W,p)\}.]$$

No upward closure and no minimal member is assumed. Dangling commitments (2.2) are inert: (W) and (W) minus a dangling member have the same accounting status. The whole family, not a selected minimal subset, is the basic object.

3.2 Contribution (unchanged)

$$[\text{CriticalBlock}(D; W, p) \iff W \in S_{_}\{E, p\} \wedge W \setminus D \notin S_{_}\{E, p\}.]$$

$$[\text{Boundary}_{_}\{E, p\} = \{(v, w) \in \mathcal{V}^2 : \text{Account}(E_{_}v, p) \neq \text{Account}(E_{_}w, p)\}.]$$

A block may contribute with no singleton contributing. Positive contribution, interference, substitution, and jointly necessary organization are all representable; the interference and infinitary examples of 1.0 are retained as permanent exhibits.

3.3 The finite monotone theorem (unchanged)

Under finiteness, upward closure, and (S) : critical membership equals the union of minimal supports (M1); indispensability from the full set equals their intersection (M2). This is conditional on monotonicity, which real support families need not satisfy.

3.4 Hardness to variation: two notions, honestly separated (R2; repairs F2)

Contract-relative preservation (the 1.0 lemma, demoted to its correct status). With organization, interpretation, variation family (V) , and job family (F) all fixed,

$$[F \text{ F}' \text{ Pres}(F') \subseteq \text{Pres}(F).]$$

This is contravariance of a fixed-preservation operator. It says nothing about any explanation's content and must not be cited as a formalization of “hard to vary.”

Intrinsic constrained variation. An account's anchoring $(_)$ at grain $_$ licenses a family of edits: those edits of the explanatory organization whose translated counterparts under $(_, _, _)$ are physically and constitutively admitted at the target, at that grain. Write this family $(V^{\wedge\{\text{int}\}}(E))$. It is generated by the account, not stipulated alongside it. Define

$$[\text{HardToVary}_{_}\ell(\mathcal{E}) \iff \text{every } v \in \mathcal{V}^{\wedge\{\text{int}\}}(\mathcal{E}) \text{ that preserves } \text{Account} \text{ preserves the active commitments' functional roles}.]$$

That is: all content-licensed ways of changing the account while keeping it an account keep the details playing their parts. The Deutsch claim — a good explanation's details are hard to vary because they play functional roles, discovered through the explanation's content and reach — is a claim about $(V^{\wedge\{\text{int}\}})$,

and is now statable: reach widens the set of questions an unchanged core accounts for (3.5), which widens the contrast evidence bearing on the anchors, which tightens ($V^{\{int\}}$). The constraint flows from the content outward, as the source intends. (H-int) is a substantive, falsifiable attribution; (H) is a bookkeeping lemma. The 1.0 error was presenting the bookkeeping as the idea.

What is not claimed. No numerical warrant follows from either notion. Splitting one feature into ten creates no knowledge. Counting preserved jobs is not a comparison between explanations with different anchors.

3.5 Reach (unchanged)

Reach is participation of an unchanged organizational core in accounts of further questions through stated anchors and additional background. The new bridge is itself content and can be originaive. No minimal-support survival is required.

Part 4 — Exact domain constructions (retained)

The following 1.0 constructions are retained without change, having passed audit: production versus identification (the pole and shadow); compatibility fibres, (I1)–(I4), including the two-balances model with kernel $((1,-1,-1))$ and identified $(b_B - b_A = 2)$; invariants and obstruction (O1) and the 23-token case; constitutive-rule queries with existence, uniqueness, ambiguity, and inapplicability as separate cases; functional and relational transport, (T1), with the one-sided/two-sided distinction; approximate transport (T2) under an explicit Lipschitz premise; and the historical-indexing rule that a changed scope creates a new claim rather than retroactively falsifying old ones.

Part 5 — Criticism, standing, evidence

5.1 Histories, active routes, situations (unchanged in substance)

A history (h) is a set of occurrences with acyclic causal precedence and a physical process interpretation; semantic support and opposition need not be acyclic. An **active route** is a connected subnetwork of actual occurrences joining a named represented input to a named operative result, satisfying the component relations and role bindings, with nonconstant dependence on the relevant represented distinction under declared contrasts. Dead branches, cancelled signals, and carrier-irrelevant details do not qualify. A situation comprises currently represented contents, active questions, uses, appraisals, attention engagements, purposes, and history links — as role relations, not mandatory databases. Attention is itself activity in the history and a possible criticism target.

5.2 Bearing and reason use (unchanged in substance)

$[\text{Bearing}(c, z, p) \iff \text{Account}(\mathcal{E}_c, p, \delta)]$

A criticism occurrence can exist without bearing. Reason use is a representation-preserving structural map from the represented objection into the response suborganization, on an active route, with content-preserving recodings preserving the transition and content changes producing the changes the system's operative deliberative rule specifies. **Using an invalid objection does not validate it.** The definition

concerns content use, not response quality. Unavailable observer data makes the attribution unresolved, not negative.

5.3 Standing (unchanged)

[Usable $_j(u) \iff \text{Lic } _j(u) \wedge \text{Scope } _j(u) \wedge \forall d \in \text{Prem}(u), \text{Live } _j(d; u).$]

Standing is enacted permission, not truth. Withdrawal of one argument leaves others and leaves the conclusion's truth untouched. Delivery of a source is not use of it. The actor can violate the coherence condition; the model then represents an error.

5.4 Elimination and tests (unchanged)

[T B I O, $\neg O \neg(T B I).$]

A failed test contradicts the conjunction. Isolating (T) requires premises about auxiliaries (B) and interpretation (I). Mechanical checks establish limited propositions under stated interpretation premises; a passing execution immunizes neither the specification nor the relevance claim against prose criticism.

5.5 Receipts (moved to appendix status; F12)

The receipt construction — leaves as event-plus-interpretation references, internal nodes as the inferences actually used, missing evidence staying missing under negation, timeout-as-delivery-evidence rather than content-evidence — is retained in Appendix A as optional descriptive mathematics. Nothing in Parts 2–8 depends on it. Its value is diagnostic: it forces an interpretation of an argument's actual dependencies. Its cost is that nothing operational follows from it, and 2.0 declines to pretend otherwise.

Part 6 — Understanding, construction, provenance

6.1 Understanding and deployability (unchanged in substance)

A system understands content (c) in a stated respect when a represented realization of (c) is integrated into its problem-directed activity, supports the declared content-governed uses, preserves role bindings and the contracted discriminations, and permits (c) to be treated as a conjectural target. Understanding a false theory is possible. $\text{Deploy}_{\{\beta, \ell\}}(s, c, \xi; U)$ requires the owned capability of 7.2; the repertoire ($R_{\{ \}, \}$ (s,)) excludes mere storage, logical consequence, and possible-later acquisition. A generic simulator is not thereby an understander; no substrate is excluded for being a simulation.

6.2 Construction and newness (unchanged in substance)

$\text{Build}_{\{\beta, \ell\}}(s, c, h, e)$ requires an actual owned or controlled subhistory with a nontrivial binding construction, not a composition of content-preserving transfers. Electricity supply, incidental ancestry, and file custody do not qualify. Learner reconstruction counts; relay does not; physically faithful copying of an operative organization is transfer, not fresh invention, where the physics permits it.

[$\text{New}(s, c, h, e) \iff \neg \exists d \in R_{\{ < e \}}(s, h), d \equiv _ \ell c.$]

Newness, acquisition, and construction can come apart. An incomplete archive makes a historical claim uncertain, not new. Initialization with a functioning organization includes that organization in the baseline.

6.3 Originative acts and episodes (unchanged)

[$\text{Origin}_{\{\beta, \ell\}}(s, c, p, h, e) \iff \text{Attempt} \wedge \text{New} \wedge \text{Build}.$]

False conjectures can be originative. A creative critical episode is a complete critical episode containing an originative contribution connected to its inquiry. Closing an episode is a decision, not a solution certificate.

Part 7 — Capability: constructor-theoretic module

7.1 Retained realization (unchanged; P2 made explicit)

[$\text{RetReal}(\pi, T, C; \chi) \iff \forall z \in C \ \forall i \in \text{dom } T \ \forall \eta \in \text{Exec}_{\Theta}(\pi, z, i; \chi),$
 η completes in $T[i]$ returning to $C.$]

Execution families must be nonempty on legitimate inputs; deadlock is not vacuous success; resources are not secretly inexhaustible. The approximate-grade version uses `Approx_0` from P2.

7.2 The retention fixed point and owned capability (unchanged)

With (F) the retention operator, $(C \text{ F}(C))$ is the invariant-witness form; (F) is monotone; the union of post-fixed sets is the greatest fixed point (CT2). $\text{Owned}_{\Omega}(\xi, \pi)$ is the physical-organizational fact of containing the control organization; $\text{Can}_{\{\Omega, \beta\}}$ requires owned retained realization or an owned finite construction history reaching one, starting from presently owned processes and admitted resources — never from an unexplained constructor for the desired achievement.

7.3 Achievement capability (unchanged)

$\text{CanAdv}_{\{\Omega, \beta\}}(\xi, p; \chi, J_p, C_I)$ (CA): for each starting configuration in the independently specified (J_p) , the maximal execution family is nonempty, every execution completes with a history satisfying the stated achievement predicate ((EK) or a stated weaker one), and the continuing organization lies in (C_I) . A first discovery is not a repeatable task; retention applies to the inquiry-enabling organization. Protocol control decisions use accessible histories only.

7.4 The possibility bridge (tone corrected; F6)

[$\text{Cap}^{\sim}_{\{q, r\}} () \text{ Admit}^{\sim}_{\{q, r\}}, \quad \text{Cap}^{\sim}_{\{\infty\}} () \text{ Poss}.$]

This records that capability implies physical possibility. It is the easy direction, its proof is one line, and 2.0 says so. No converse is claimed: preservation roles without criticism (biology) and criticism without preservation (a lost discovery) both exist. The constructor-theoretic knowledge predicate $\text{CTK}_{\{\emptyset, \mathcal{E}\}}$ and its conditional bridge to (EK) are retained with the realization ecology (E) displayed. The bridge is conjectural in both directions and labeled as such.

Part 8 — Recursion and universality

8.1 Recursive critical capacity (unchanged)

Scrutinizability of an aspect (d): an owned, physically admitted continuation in which a description of (d) becomes a represented target, conjecture and criticism can be directed at it, and results can return to its operative use — with at least one alternative subsidiary result permitting a different use-state, unless the contract establishes a single applicable use. Recursive critical capacity is closure under every finite admitted target chain:

[$n < \infty$ admitted chains of length n , an owned enabling continuation.]

No completed infinite regress, no infinite simultaneous storage, and no internal perfect truth predicate is required. Chains concerning the evaluator itself are not silently deleted.

8.2 Barriers and extension (unchanged)

An explanatory barrier is an independently characterized domain blocking the relevant owned capability under every admitted, non-question-begging enabling condition. Frozen components are not automatically barriers; editable parameters are not automatically competence. A historical domain-extension event is defined relative to an independently specified (D), proves nothing about every subject, and is not required of a universally capable system.

8.3 Admissible enabling conditions, formalized (R3; repairs F1/F3)

An enabling condition for task (T) is **admissible for a capacity claim** only if:

1. it is described by physically and semantically meaningful resources and contributions, independently of the success claim (1.0 rule, retained);
2. it does not deliver the target achievement itself — for an understanding task (U_c), must not include delivery of (c)’s operative organization to the recipient in a form requiring no binding construction (complete exposition that still requires reconstruction *is* admissible for understanding; delivery of a functioning realization is not);
3. it does not include a competent replacement thinker or a complete functioning solution for an origination task.

Condition 2 is the formal repair. “Understanding given arbitrary help” is a weaker predicate, available as (U_c)-with- -explicit, and must never be silently strengthened to the admissible-quantified form.

8.4 The universal class (indices threaded; F10)

With grains and boundaries fixed throughout (9.4):

[$UU \{ , , \} (s,) \quad c \in \Theta \exists \chi [\text{Admissible}(\chi, U_c) \wedge \text{Enable}(s, U_c, \chi) \wedge \text{Can_}\{\Omega, \beta\}(\xi_0, U_c; \chi)],]$

[$UC \{ , , \} (s,) \quad p \in P \sim \{adv\} \Theta \exists \chi [\text{Admissible}(\chi, A_p) \wedge \text{Enable}(s, A_p, \chi) \wedge \text{CanAdv_}\{\Omega, \beta\}(\xi_0, p; \chi, J_p, C_I)],]$

[$UECS = \{ (M, s, , ,) : M \in RC \wedge UU \wedge UC \}.]$

The domains come from P6, specified independently of the candidate’s record. The same $\Omega, ,$ hold across all quantifiers. Universality demands neither that every attempt succeed nor simultaneous possession of all contents. The qualitative-jump conjecture remains compatible and unproved; equality of “has once extended itself” with (U3) is not adopted.

Honest status note. With admissibility now formalized, (U1) is stronger than 1.0's version but remains existentially quantified over a physically meaningful for each content. Whether any physically realized system satisfies (U3) is unknown in both directions; the class is defined so that the question is well-posed, not so that it is answered.

Part 9 — Repair, knowledge creation, and the class

9.1 Repair (unchanged in substance)

With claimed repair obligations (O) and protected obligations (P) fixed for a comparison:

[Repair $_ \{O, P\}(\xi, \xi'; \Delta) \iff \exists o \in O[\neg o(\xi) \wedge o(\xi')] \wedge \forall r \in P[r(\xi) \Rightarrow r(\xi')] \wedge \text{ProducedBy}(\Delta, \xi, \xi'; O).$]

Local, attributable, non-ranking. Losses outside (P) must be exposed. Adding an account while losing a protected one is not repair. Replacing a false identification with a correct account of underdetermination can be.

9.2 Created explanatory knowledge (notation repaired; F9)

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$$\begin{aligned} \text{CreateEK}_{\beta, \ell, O, P}(s, \Delta, h, e) \iff & \text{CreativeCriticalEpisode}(s, \Delta, h, e) \wedge \text{Repair}_{O, P}(\xi, \xi'; \Delta) \\ & \wedge \exists o \in O_{\text{ep}} \exists c, p_c, e_c [e_c \preceq_h e \wedge \neg o(\xi) \wedge o(\xi') \\ & \wedge \text{Origin}(s, c, p_c, h, e_c) \wedge \text{Account}(\mathcal{E}_c, p_c) \\ & \wedge c \in \text{Result}(\Delta) \wedge \text{Deploy}_{\beta, \ell}(s, c, \xi'; U_c) \wedge \text{ProducesVia}(\Delta, c, o; \xi, \xi')]. \end{aligned}$$

]

(E_c) is the structural account carried by content (c) — the reading 1.0 used without introducing. Factive about the claimed repair; fallible throughout; historical, so later loss does not retroactively uncreate. A stored artifact for a hypothetical reader is not possession.

9.3 No merit function (unchanged)

No endorsement count, survival count, or feature count enters (G), (P), or (EK) as a warrant. Particular reasons for choices are substantive and criticizable; a universal order on alternatives is not supplied and not missed.

9.4 The grain-fixing discipline (R6; repairs F8)

For any attribution under (E), (E-), (G), (EK), or (U1)–(U3):

1. The grain , boundary , and contrast contract (C) must be fixed **before** the contrast evidence bearing on the attribution is examined.
2. Changing or mid-assessment resets the attribution: the new assessment is a new claim with its own registry entries (R1-1), and may not cite evidence gathered under the old grain without re-examination at the new one.
3. A counterexample may not be absorbed by re-graining after the fact. A proposed re-grain that dissolves a counterexample is itself a new constitutive proposal and must be defended as such.

This rule does not make the constitutive conjecture falsifiable in the Popperian strong sense — conceptual analyses never are — but it removes the framework’s one structural escape hatch, at the price of making applications slower. That price is accepted deliberately.

9.5 The class collected (indices threaded)

Primitive data: Θ (P2), histories, organizations satisfying (O), contents and occurrences with **Rep** (P1), question contracts, control and appraisal histories, continuity Ω , normative relations where invoked (P4), domains (P6), grains and boundaries (P7). Base class: interpretations supplying these data, typed, satisfying physical-realization and reference-coherence conditions where physical attributions are made. Then the creative-episode class (an instance of (G) in a critical episode), the knowledge-creation class (an instance of (EK)), the recursive-capacity class ((RC)), the universal class (U3). An originaive event certifies nothing about universal capacity; a small creative act is not a certificate of personhood, consciousness, or moral standing.

Part 10 — Positioning and defeat conditions

10.1 Positioning map (R8)

ECS 2.0’s components have near-neighbors it does not compete with and should be read alongside: structural-causal explanation (Pearl, Halpern — cited in 1.0 as R6); interventionist accounts of explanation (Woodward), whose “invariance under intervention” is a special case of (F) over an intervention contract; mechanistic explanation (Craver, Bechtel), whose how-actually requirement is the anchoring clause’s kind-typing; bisimulation and simulation in process algebra (relational transport, (T1)); identifiability theory ((I1)–(I4)); sufficiency in statistics (the factorization shape of (I2)); greatest fixed points in coalgebraic process theory ((CT2)); constructor theory and its process-theoretic formalization (R2–R5). The framework’s claim is not priority over any of these. Its claim is that the *assembly* — typed restriction, anchoring with exposed kind-typing, graded accounting with exposed residue, provenance-sensitive construction, admissible enabling, and grain discipline — states satisfaction conditions for explanatory-attribution claims that none of the neighbors states in full, because none of them is trying to.

10.2 What would defeat 2.0

1. A candidate satisfying (E) at a grain fixed per 9.4, with named registry supplies unfalsified, that is nevertheless no explanation — refutes sufficiency.
2. A genuine explanation unsatisfiable under (E-) for any at any pre-fixed grain — refutes necessity. Non-causal mathematical explanation and aesthetic explanation are the expected stress cases; 2.0’s treatment of both is explicitly provisional.
3. A real inquiry whose commitment structure cannot be typed by the restriction algebra of 2.2 — restricts (S)–(M2) to better-behaved domains.
4. A case where the intrinsic variation family (V^{int}) cannot be determined from the anchoring — restricts (H-int), and would show the repaired hard-to-vary notion to be inapplicable exactly where it matters.
5. A demonstration that every admissible supply of P1 (representation) is itself either circular or empirically empty — would collapse the framework’s applicability, though not its consistency.
6. A mathematical counterexample to any displayed theorem under its stated assumptions — exposes an error; none is known after audit.

What does *not* defeat 2.0: showing that an application fails (that restricts the application); showing that membership is undecidable in a presentation (nothing here claims decidability); showing that a frozen implementation or an uncooperative archive blocks a particular attribution (that is a barrier claim requiring its own argument, per 8.2).

10.3 What 2.0 is for

The realistic function of this framework, stated once and not inflated: it is a **contract for attribution claims**. Anyone asserting that a system explained, used a reason, understood, created knowledge, or is universal, owes the registry entries, the grain, the residue, the active-route witness, and the provenance account that the relevant predicate requires. Anyone auditing such a claim can use its un-supplied conjuncts as a checklist of exactly what has not been shown. Used that way, the framework is unfalsifiable in the harmless way checklists are; its constitutive conjecture — that these conditions are also *correct as an analysis of explanation* — remains open, stressed at the points listed in 10.2, and is the place to attack.

Appendix A — Evidence receipts (optional descriptive mathematics; F12)

[Retained verbatim in substance from 1.0: evidence leaves as event-plus-interpretation references; derivation trees with the inferences actually used; distinct sufficient routes as distinct receipts; conjunction and disjunction discipline; negation exchanging positive and negative receipt sets while missing evidence stays missing; eligibility separating completed-answer evidence from delivery failure. No downstream definition in 2.0 depends on this appendix.]

Appendix B — Mathematical checking for this document

2.0 introduces new mathematics only in 2.2 (typed restriction), 2.3 (graded accounting), 3.4 ((H-int)), 8.3 (admissibility), and 9.4 (grain discipline). Of these, 2.3, 8.3, and 9.4 are definitional disciplines with no theorem content; 2.2 is a typed operation family whose claimed property is well-formedness by construction; (H-int) is a definition, not a theorem. All retained theorems were verified in the 1.0 audit (ECS_1_0_Audit.md, Part 1), including independent recomputation of the monotone-family census (4, 18, 166 families; 1, 11, 151 union/intersection disagreements — reproduced exactly). No proof assistant checked this manuscript. No experiment establishes that any actual system satisfies any class defined here.

Appendix C — Change log from 1.0

Change	Motivation
Part 0 Primitive Load Registry added	F1, F7 / R1
Restriction made a typed algebra with dangling-reference and map-truncation rules	F4 / R4
Graded accounting (E-) with exposed residue added	F5 / R5

Table 2 – continued

Change	Motivation
(H) demoted to bookkeeping lemma; (H-int) intrinsic hardness added	F2 / R2
Admissible enabling conditions formalized; (U1) strengthened	F3 / R3
Grain-fixing discipline added	F8 / R6
(EK) notation repaired; / threaded through Parts 8–9	F9, F10 / R7
Framework retitled; receipts moved to appendix	F11, F12 / R7
Positioning map added	F13 / R8
CT-bridge tone corrected	F6
All other content retained from 1.0	Audit Part 3